

Proxy-Lag Cascade Ensemble: A Multi-Horizon Architecture for EV Charging Utilization Forecasting

Xuan-Thanh Trinh^{1*}

^{1*}School of Electrical Engineering, Hanoi University of Science and Technology, Hanoi, Vietnam.

Corresponding author(s). E-mail(s): thanh.tx@hust.edu.vn;

Abstract

The rapid expansion of Electric Vehicle (EV) charging infrastructure introduces severe non-linear fluctuations in grid demand, necessitating highly accurate utilization forecasting models. Traditional forecasting methods often fail to capture abrupt load peaks or succumb to phase lag when attempting multi-horizon predictions. In this paper, we propose a novel Proxy-Lag Cascade Ensemble architecture inspired by Model Predictive Control (MPC) principles. The architecture consists of a 24-hour Rule-Based Two-Stage trajectory model and a 1-hour Residual Corrector model. By employing Target Power Transformation (TPT) alongside a mathematically formulated Custom Peak Loss function, our methodology effectively captures extreme peak behaviors while strictly avoiding data leakage. We demonstrate that tree-based tabular models, specifically XGBoost, achieve a state-of-the-art RMSE of 0.0664, surpassing both deep learning approaches and traditional time-series baselines. The robust theoretical alignment between the macro-level predictive trajectory and micro-level residual correction establishes a new standard for EV demand forecasting, ensuring stability for modern smart grids.

Keywords: Electric Vehicle, Load Forecasting, XGBoost, Residual Boosting, Phase Lag, Two-Stage Architecture

1 Introduction

2 Material and Methodology

2.1 Data Preprocessing and Feature Engineering

2.2 Data Characteristic Analysis and Algorithm Selection

2.3 The Mathematics of Peak Catching: Custom Loss and Target Power Transformation (TPT)

2.4 Overcoming the Valley: The Two-Stage Rule-Based Architecture

2.5 Compensation Thinking: The 24h Horizon Limitation

2.6 Building the 1h Corrector Layer: From Failed Attempts to the Proxy-Lag Cascade

3 Results and Discussion

3.1 Superiority of Tabular Models over Deep Learning

3.2 Performance of Peak-Catching Mathematics and Two-Stage Architecture

3.3 Breaking Phase Lag with Residual and Proxy-Lag Integration

3.4 State-of-the-Art Performance and Deployment Discussion

4 Conclusion

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

Please refer to Journal-level guidance for any specific requirements.

Acknowledgements. Acknowledgements are not compulsory. Where included they should be brief. Grant or contribution numbers may be acknowledged.

Please refer to Journal-level guidance for any specific requirements.

Declarations

Some journals require declarations to be submitted in a standardised format. Please check the Instructions for Authors of the journal to which you are submitting to see if

you need to complete this section. If yes, your manuscript must contain the following sections under the heading ‘Declarations’:

- Funding
- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use)
- Ethics approval and consent to participate
- Consent for publication
- Data availability
- Materials availability
- Code availability
- Author contribution

If any of the sections are not relevant to your manuscript, please include the heading and write ‘Not applicable’ for that section.

Editorial Policies for:

Springer journals and proceedings: <https://www.springer.com/gp/editorial-policies>

Nature Portfolio journals: <https://www.nature.com/nature-research/editorial-policies>

Scientific Reports: <https://www.nature.com/srep/journal-policies/editorial-policies>

BMC journals: <https://www.biomedcentral.com/getpublished/editorial-policies>

Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.